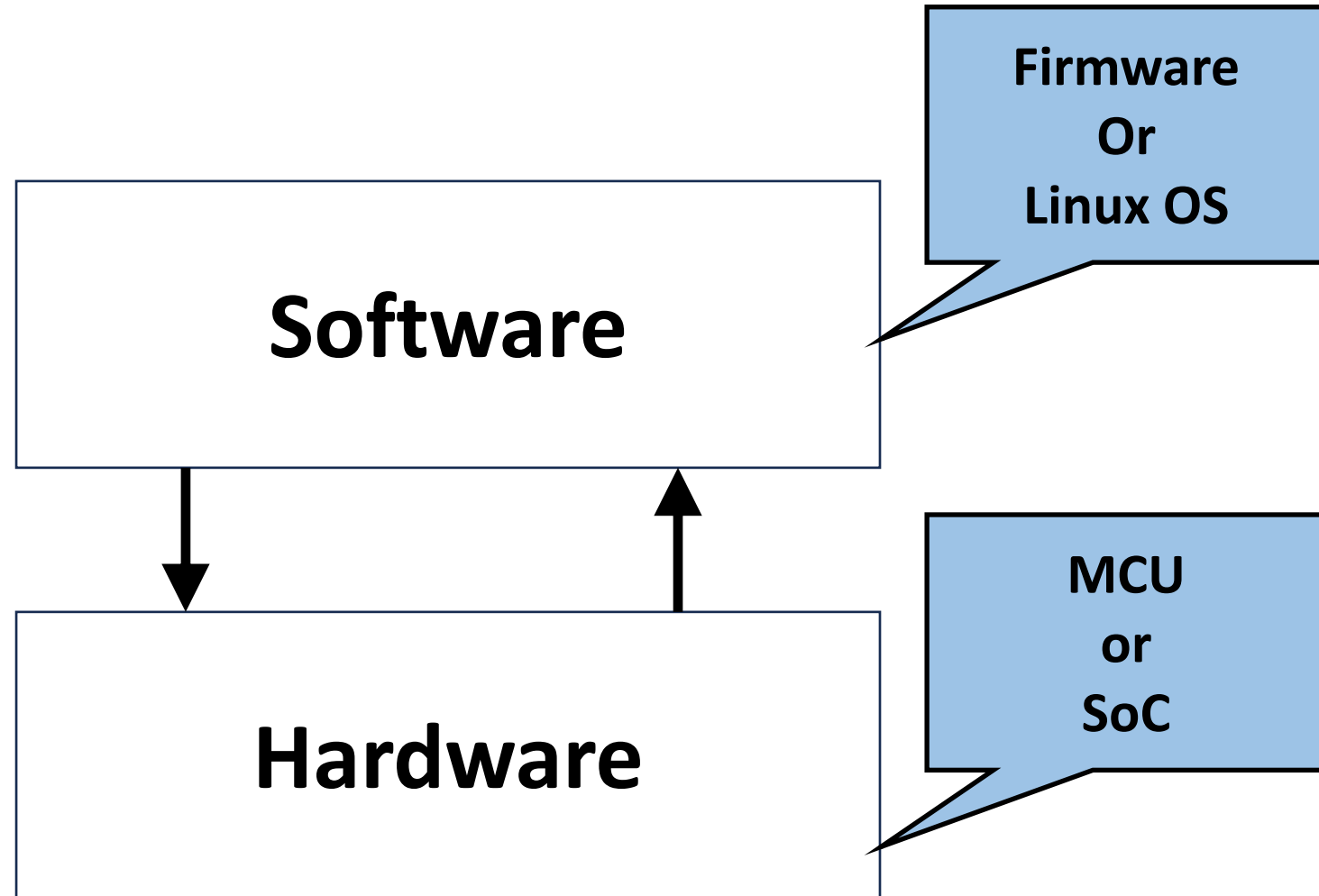


Introduction to Linux Device Driver

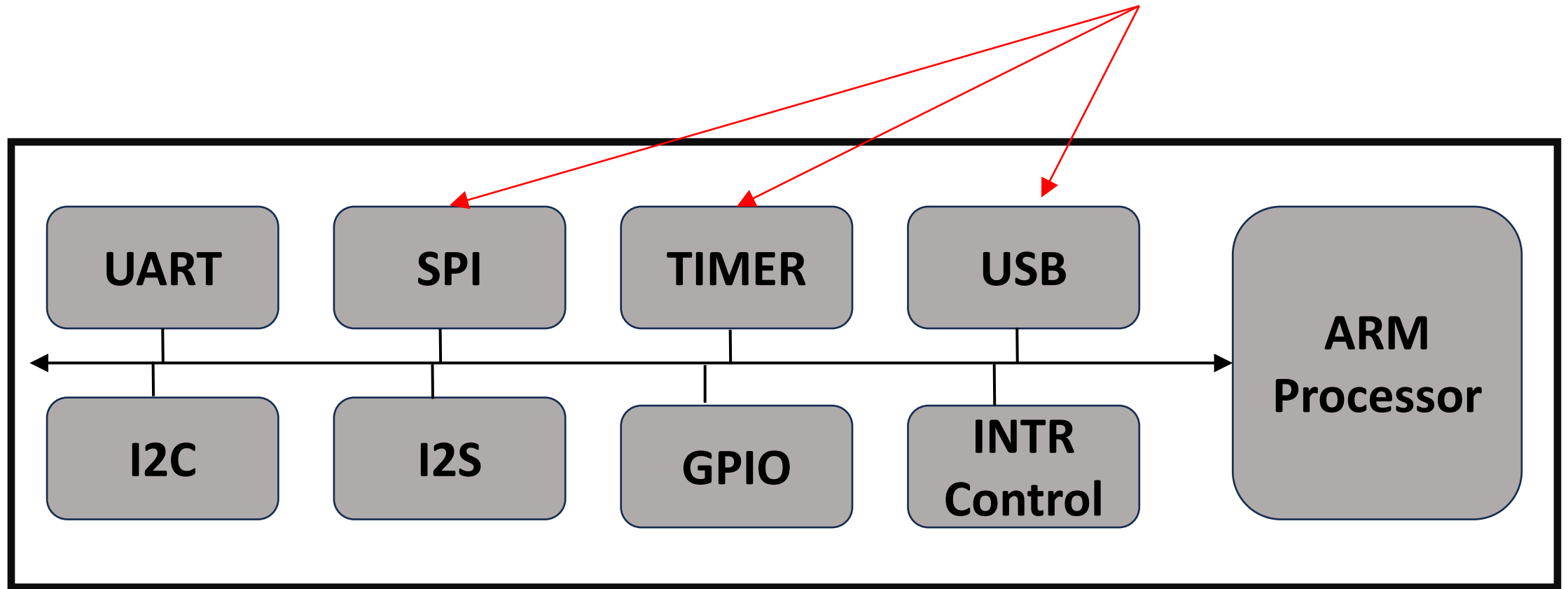
What is Device Driver in Linux?

- A **device driver** is a piece of software that allows the Linux kernel to communicate with hardware devices in a standardized and secure way.
- It acts as an interface between the operating system and hardware components like keyboards, mouse, storage devices, network cards, and embedded peripherals.

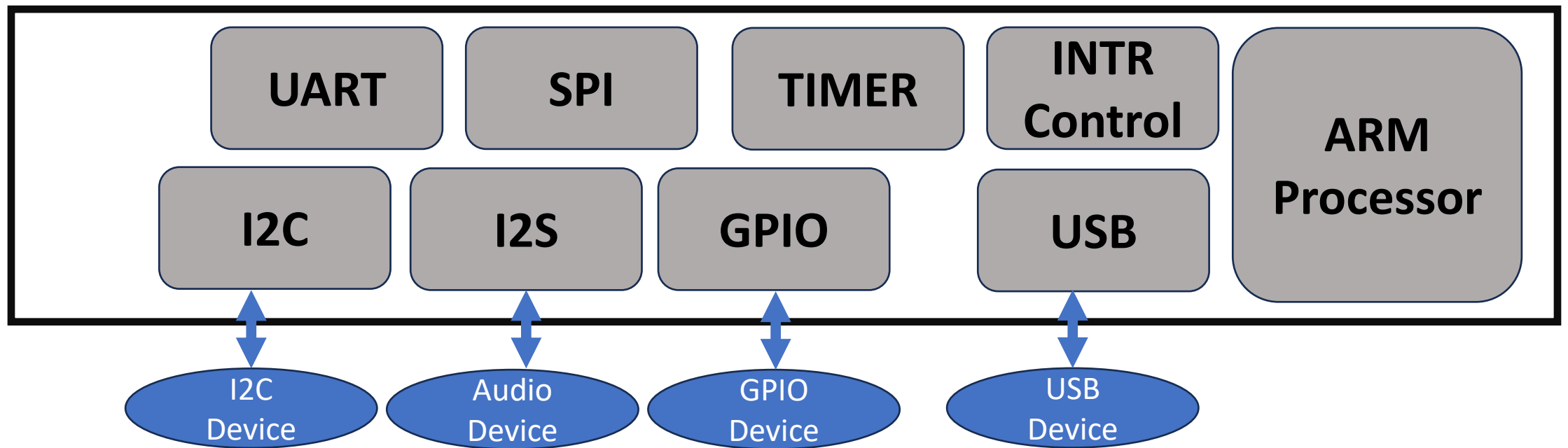


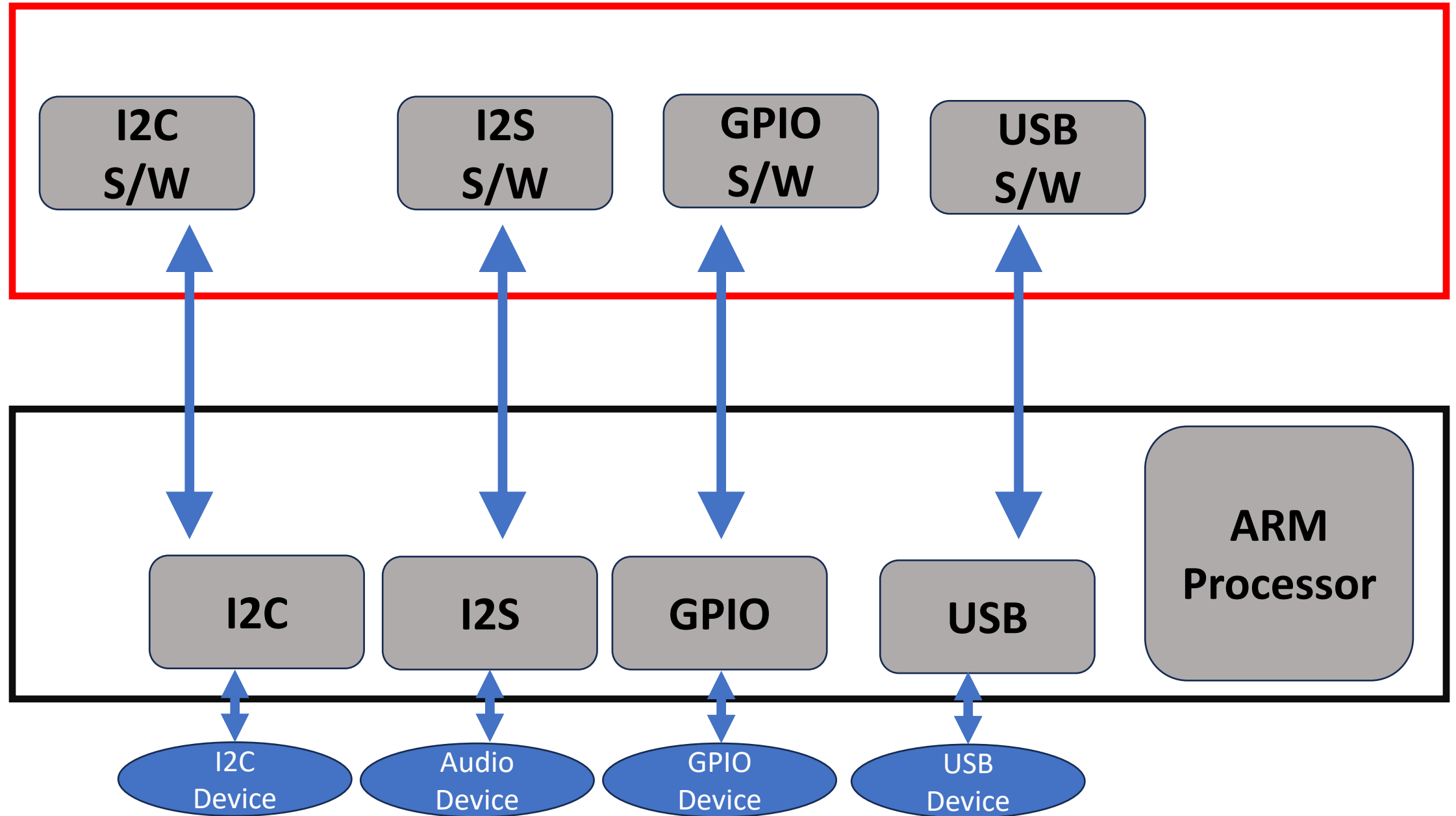
SoC Internals

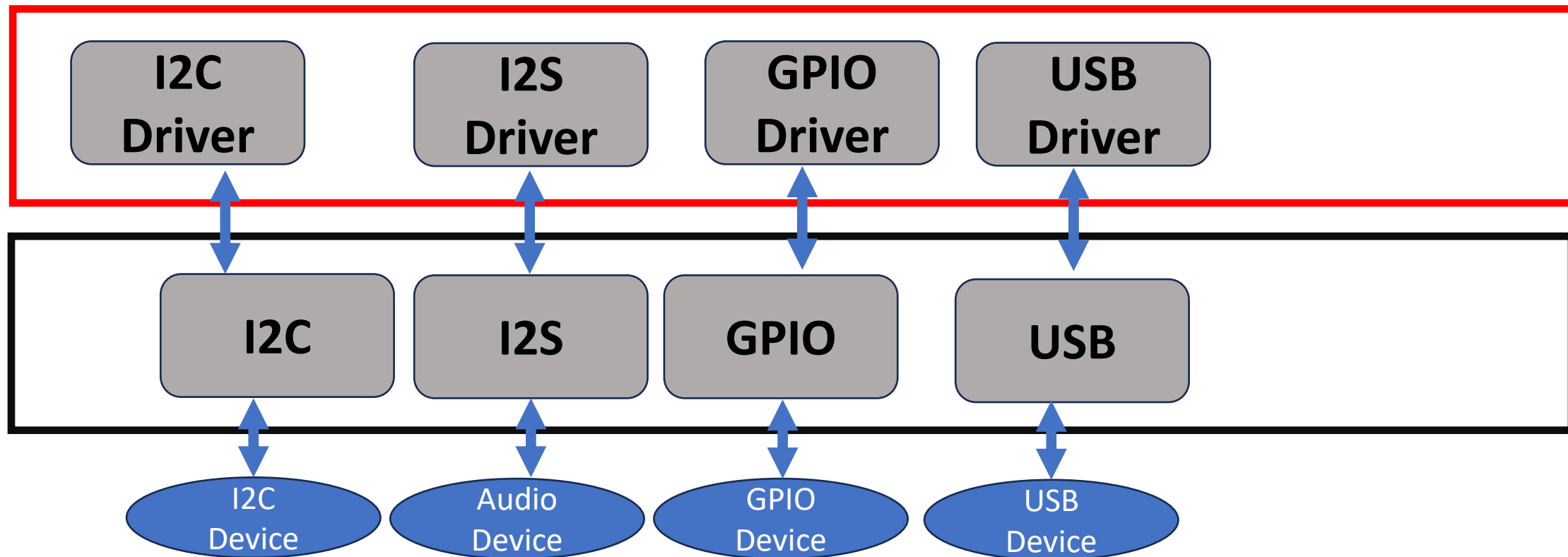
SoC peripherals

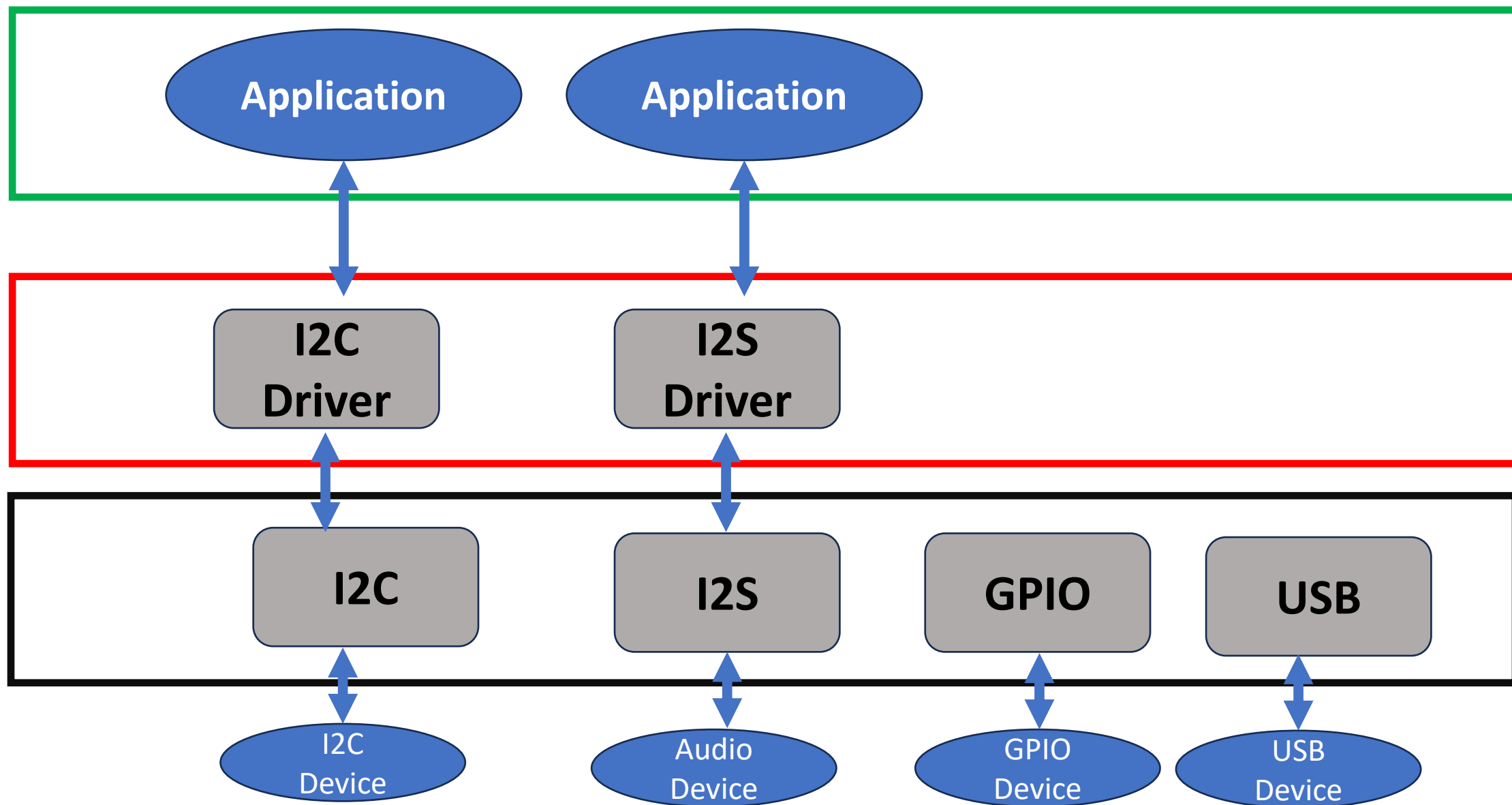


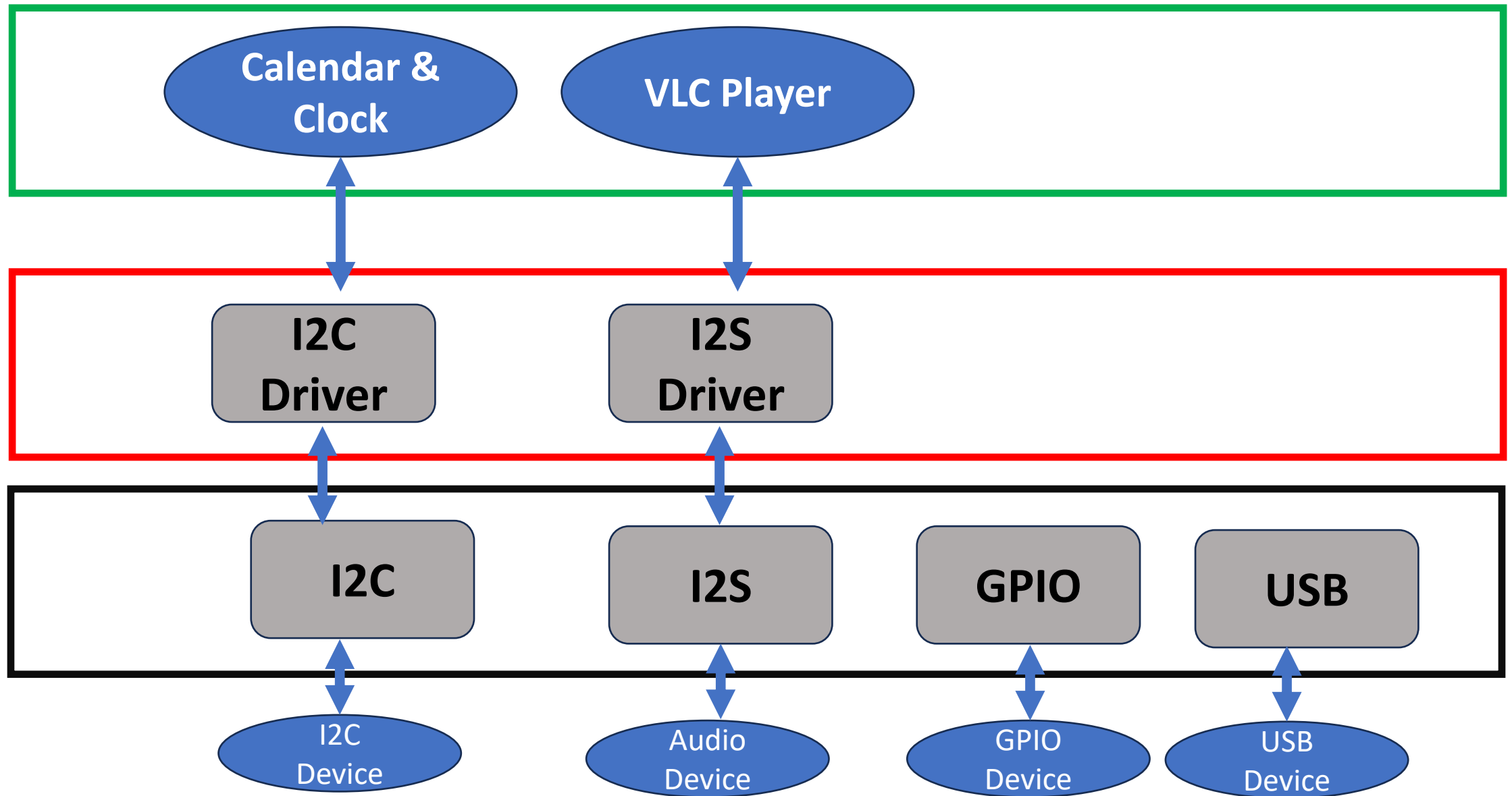
Any SoC

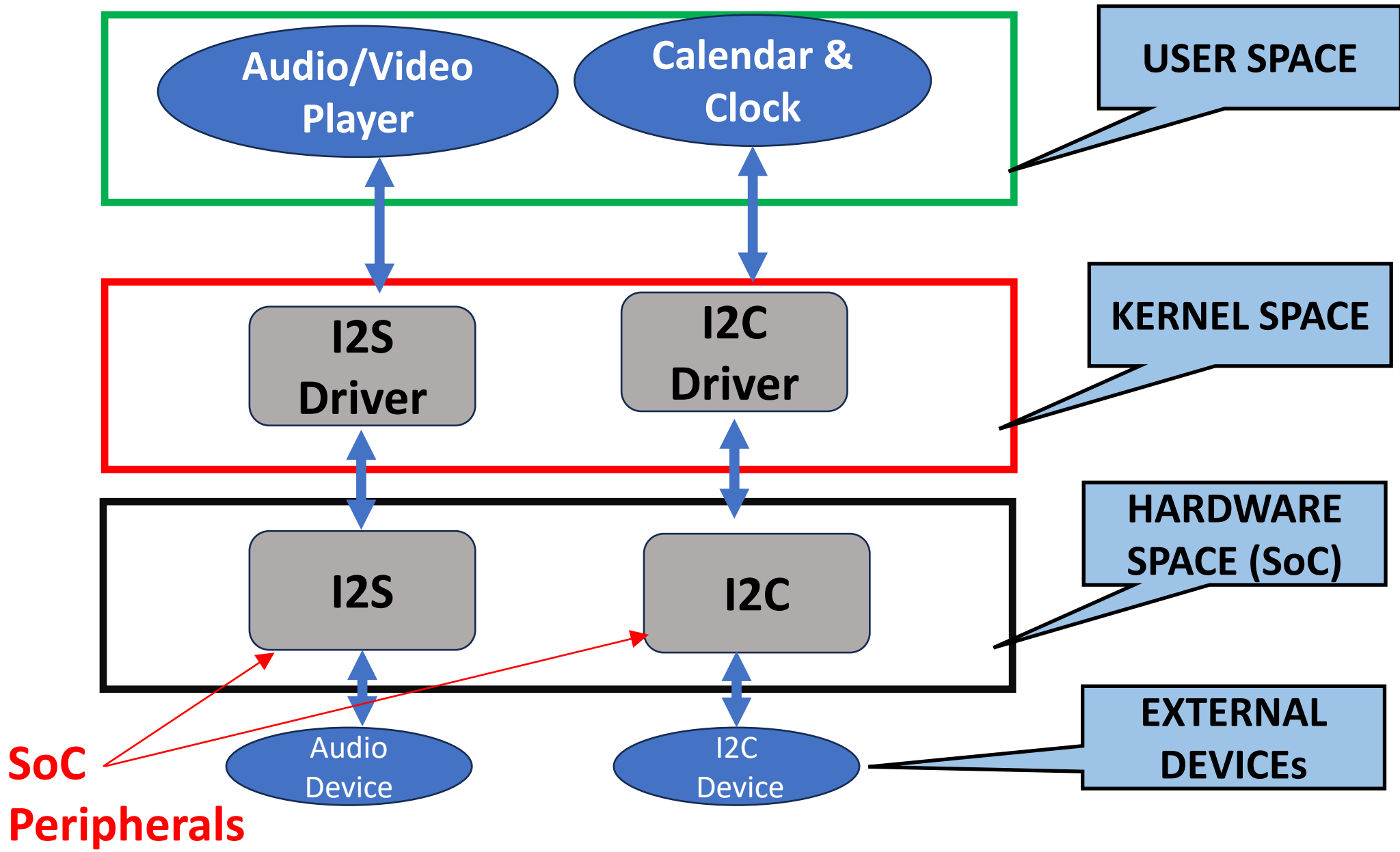










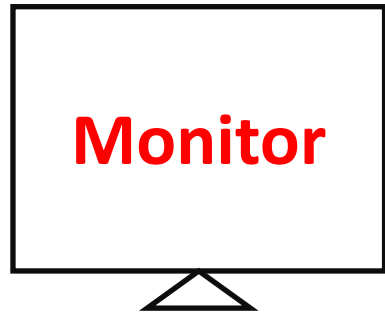


Why device drivers are required in OS?

1. Bridging the Gap Between New Hardware and Kernel.
2. Enabling OS Portability Across Platforms.
3. Supporting New Hardware Features.
4. Ensuring Forward and Backward Compatibility.

Classification of devices in Linux

- In Linux and other UNIX-like operating systems, devices are categorized based on how they handle data. These devices are represented as device files (typically found in /dev) and are classified into:
 1. Character devices (byte oriented)
 2. Block devices (block oriented)
 3. Network device (packet oriented)



List of computer peripherals

CAMERA

GPIO

KEYBOARD

MOUSE

RTC

TOUCHSCREEN

SOUND CARD

SERIAL PORT

HDD

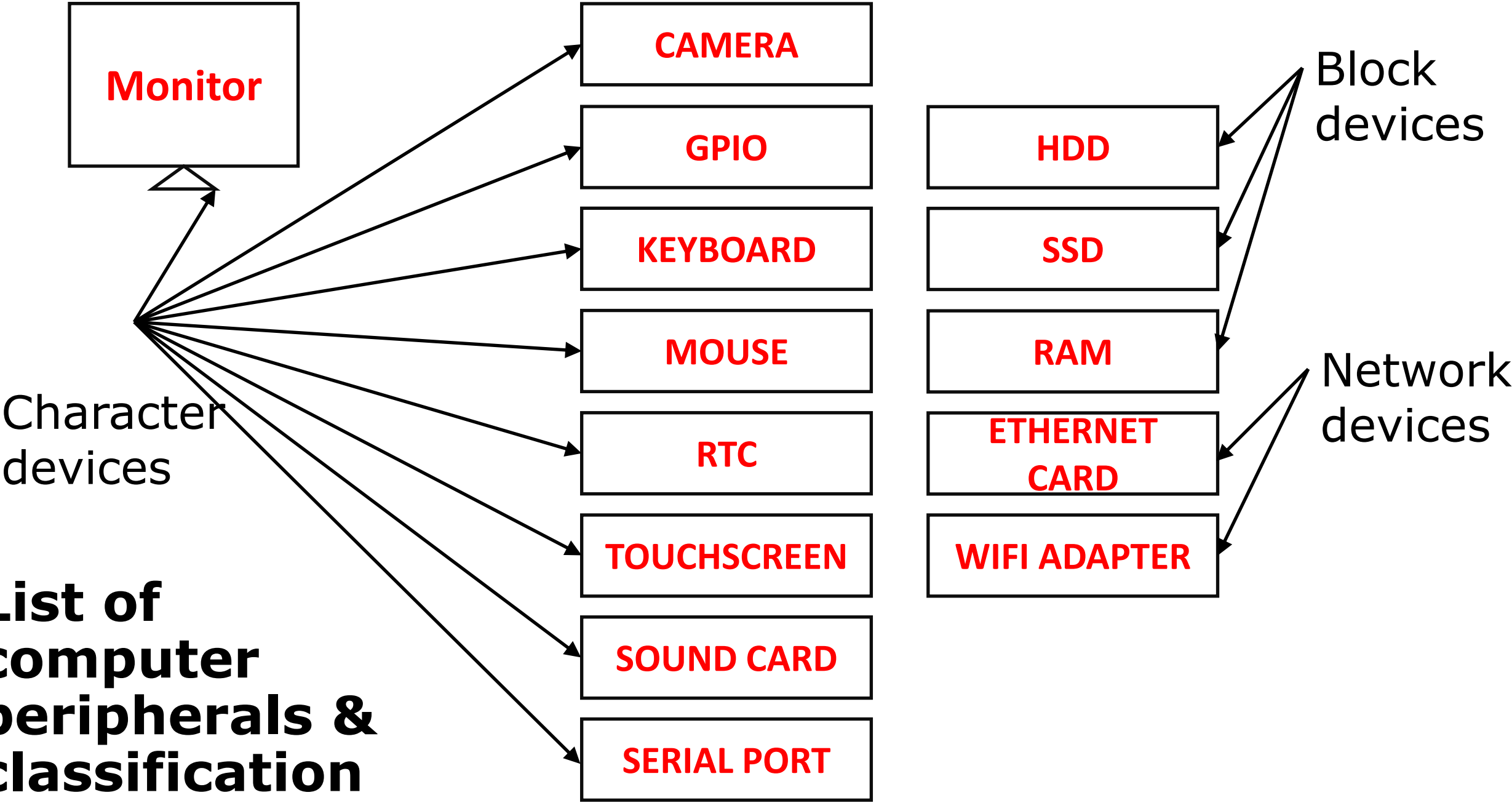
SSD

RAM

**ETHERNET
CARD**

WIFI ADAPTER

**List of
computer
peripherals &
classification**



Types of Device Drivers

1. Character drivers (byte oriented)

These devices transmit & receive data byte by byte (character by character), unlike block devices that handle data in fixed-size chunks.

2. Block drivers (block oriented)

These devices transmit & receive data in fixed-size chunks/blocks.

3. Network drivers (packet oriented)

These devices transmit & receive data in packets.

List of Character devices in Linux

- **Serial & UART Devices** (RS-232, UART, USB-to-Serial converters)
- **Input Devices** (Keyboards, Mice, Touchscreens, Joysticks)
- **GPIO** (Raspberry Pi GPIO, BeagleBone, Embedded controllers)
- **I2C** (EEPROMs, Temperature Sensors, RTC)
- **SPI** (Flash Memory, ADC/DAC, Display Controllers)
- **CAN** (Automotive ECUs, Industrial Automation)
- **Framebuffer** (LCDs, OLEDs, Embedded Screens)
- **Sound & Audio Devices** (Microphones, Speakers, Sound Cards)

List of Block devices in Linux

- Hard Disk Drives (HDD) & Solid State Drives (SSD)
- External Storage
- Memory Cards (SD, microSD)
- RAM Disks
- Optical Drives (CD/DVD/Blu-ray)

List of Network devices in Linux

- **Ethernet** (LAN Cards, PCIe Ethernet Controllers, USB-to-Ethernet Adapters)
- **Wi-Fi Adapters** (PCIe/USB Wi-Fi Adapters, Built-in Wi-Fi Modules)

What is Linux?

- Linux is an open-source, OS kernel.

What is Linux Operating System?

- A Linux distributions (or Linux distro) is a complete operating system built around the Linux kernel (the core part of the OS) and bundled with a set of software, tools, graphical interface and a package management system.

Major Linux distributors

1. Red Hat, Inc.

- Famous for **Red Hat Enterprise Linux (RHEL)** and **Fedora**.
- Focus: Enterprise servers, cloud, business solutions.



2. Canonical Ltd.

- Creator of **Ubuntu**.
- Focus: Desktop users, cloud services, IoT (Internet of Things).



3. SUSE

- Creator of **SUSE Linux Enterprise Server (SLES)** and **openSUSE**.
- Focus: Enterprise servers, cloud infrastructure, stability.



Key Features of Linux

- **Open Source** – Anyone can view, modify, and distribute the source code.
- **Multitasking & Multiuser** – Supports running multiple programs and users simultaneously.
- **Stability & Security** – Known for its robustness and strong security features.
- **Customization** – Highly customizable with different desktop environments (GNOME, KDE, XFCE) and package managers.
- **Command-Line Power** – Offers powerful terminal commands for system management.

Writing your First Linux driver

```
/hello.c/
#include <linux/module.h>
#include <linux/version.h>
#include <linux/kernel.h>

static int _init hello_init(void){
    printk(KERN_INFO "Namaskar\n");
    return 0;
}

static void _exit hello_exit(void){
    printk(KERN_INFO "Bye Kernel Space\n");
}

module_init(hello_init);
module_exit(hello_exit);
MODULE_LICENSE( "GPL");
MODULE_AUTHOR("pramod <email: pramod@vectorindia.org>");
MODULE_DESCRIPTION("Our First Driver");
```

```
#Makefile

obj-m +=hello.o

all:

    make -c /lib/modules/$(shell uname -r)/build M=$(PWD) modules

clean:

    make -c /lib/modules/$(shell uname -r)/build M=$(PWD) clean
```

Commands to remember

1. 'make' : to build your linux driver (kernel module)
2. 'make clean' : to clean extra created during build files from your dir
3. 'insmod' : to load a kernel module (linux driver) to kernel space dynamically.
4. 'rmmod' : to unload kernel module from kernel space dynamically.
5. 'modinfo' : module information
6. 'lsmod' : list of recently loaded modules in kernel space